



Mingming Cai

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🌐 [Mingming's LinkedIn](#)

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Biography

- Ph.D. candidate in sustainable transportation and human-centered AI, with a concurrent M.S. in Statistics from UW.
- Academic journey centered on applying Statistical, Machine Learning, and Deep Learning models to travel demand prediction | Industry experience with state-of-the-art Generative AI models
- Proficient in Python (5+ years), R (5+ years), and SQL (3+ years), with dedicated experience in software system design using C++ and C# (1+ years), distributed computing using MapReduce and Spark (2+ years), Git (2+ years), Linux (2+ year), and AWS (1+ year).
- Passionate about state-of-art and responsible AI techniques (Robustness, Fairness) to promote human-centered AI services.
- A collaborative person with effective communication and contributions to diverse projects within interdisciplinary teams.

Education

University of Washington

September 2020 - May 2025 (Expected)

Ph.D. in Transportation Economics and Regional Planning (GPA: 3.9 / 4.0) - Minor: Data Science

Seattle, WA

- Relevant Coursework: Machine Learning for Big Data (Recommendation System, NLP, Social-Network Graph Analysis), Statistical Modeling & Prediction & Computing (TensorFlow, PyTorch), Time Series Analysis, Statistical Analysis of Social Networks

University of Washington

September 2023 - May 2025 (Expected)

M.S. in Statistics (GPA: 3.8 / 4.0) - Minor: Machine Learning

Seattle, WA

- Relevant Coursework: Causal Modeling, Bayesian Statistics, Design and Analysis of Experiments

Wuhan University

September 2017 - June 2020

M.S. in Land Resource Management (GPA: 3.9 / 4.0)

Wuhan, China

- Relevant Coursework: Python Programming, Database System (SQL), Data Visualization (D3.js), Operational Research

Wuhan University

September 2013 - June 2017

B.E. in Natural & Land Resources Management (GPA: 3.8 / 4.0)

Wuhan, China

- Relevant Coursework: Advanced Mathematics, Linear Algebra, Probability Theory and Statistics, Database Principles (SQL, Oracle), Programming Language Design (C), Software System Design (C, C++), Microeconomics, Macroeconomics, Data Structures and Algorithms, Systems Programming (C, C++)

Research Projects

Predicting Ride-hailing Demand in Low-density Areas: A Probabilistic Multi-task GNN Approach | *Leading Researcher*

Funded by Washington State Department of Transportation, ongoing

- Proposed a probabilistic multi-task framework that enhances the generalization capability of GNNs in predicting sparse ride-hailing demand in low-density areas, through the integration of probabilistic modeling and sociodemographic features.
- Demonstrated that GNNs incorporating the proposed framework can outperform other fundamental statistical, ML, and DL models (XGBoost, GCNs, GraphSAGE).

Airbnb Recommender System: NLP-Driven Semantic Information Mining for User Preferences | *Leading Researcher*

Funded by Paul G. Allen School of Computer Science & Engineering, University of Washington, 2023

- Implemented end-to-end sentiment analysis (fine-tuned BERT), topic modeling (developed and fine-tuned BERTopic, LDA), and multi-task learning (using PyTorch) to extract semantic information from millions of user review comments and develop personalized recommendations, countering popularity bias and cold-start problem.
- Efficiently preprocessed large textual data (tokenization, stopwords removal, lemmatization, etc.) by utilizing NLTK & scikit-learn.
- Tuned hyperparameters via coherence analysis and utilized Pyspark and Google Cloud to accelerate the model training process for the large dataset.

Time-series Forecasting of Real Estate Taxes using Deep Learning Techniques | *Leading Researcher*

Funded by National Natural Science Foundation of China, 2017

- Developed time-series prediction models using ARIMA, exponential smoothing models, and recurrent neural networks to forecast real estate tax revenue for Shanghai pilot projects, achieving 72.9% accuracy with RNN and 5%-12% lower with the other models.
- Enhanced data pipeline and feature engineering via variable transformation, missing data imputation, and input standardization.
- Led the research and published research findings in an academic journal.

Work Experience

Data Scientist Intern

September 2024 - December 2024

Amazon

Sunnyvale, CA

- Defined the research problems and designed the scientific solutions for the performance improvement on RAG system in enhancing LLM capabilities of responding to user queries.
- Applied and extended state-of-the-art information retrieval techniques to improve LLM inference in generating robot responses, achieving an increase in Recall@5/10 for context retrieval and a decrease in latency by over 50 % reduced tokens.
- Converted the methodological innovation design into Python scripts, tested it independently, and delivered it in collaboration with software engineers.

Data Analyst - Modeling Intern

June 2023 - August 2023

Chicago Metropolitan Agency for Planning

Chicago, IL

- Optimized EMME auto-routing algorithm for simulation and cross-validated it with Google Maps routing, reducing the percentage of highly mismatched routes by 70% and moderately mismatched routes by 45%.
- Visualized the route validation results using ArcGIS and R, deploying interactive online maps.
- Developed Python scripts to efficiently measure transit accessibility and transportation greenhouse gas emissions, integrating data inputs from multiple sources.

Research Scientist Intern - ML Modeling

June 2020 - August 2020

Wuhan Transportation Development Strategy Institute

Wuhan, China

- Implemented Conv-LSTM neural networks to predict citywide dockless bike sharing demand, capturing spatio-temporal dependencies in the demand and improving overall prediction accuracy (MAPE) by 5%-15% compared to ARIMA, LSTM, GRU.
- Developed performance-based stacked tree-based models (XGBoost, LightGBM, CatBoost) to predict ride-hailing travel time using city-scale mobility data, achieving a MAPE of 3%-18% for off-peak and peak predictions.
- Collected open data using web scraping techniques in Python (Beautiful Soup, Scrapy), acquiring real-time data of transit services.

Data Science and Big Data Analytics Intern

June 2018 - August 2018

Wuhan Geomatics Institute

Wuhan, China

- Developed spatio-temporal data mining algorithms to identify short-term and long-term citywide commute behavior patterns using cell phone signaling data, informing government decision-making on transportation investments.
- Enhanced data visualization using Vega and D3.js. Published in the 2019 annual report of the Wuhan Municipal Government.

Certificates & Awards

- 2024 Outstanding PhD Student Award for Academic Achievement and Leadership, University of Washington
- 2022 Special Student Service and Contribution, University of Washington
- 2019 Grand Prize, Geographic Information Science and Technology Innovation Contest, Ministry of Natural Resources of China
- 2018 First-level Scholarships, Wuhan University (Top 5 %)
- 2017 Outstanding Undergraduate Thesis, Department of Education of Hubei Province, China
- 2016 Honorable Mention, Interdisciplinary Contest in Modeling , Consortium for Mathematics and its Applications (COMAP)
- 2015 National Scholarship, Ministry of Education of China (Top 1 %)
- 2014 National Scholarship, Ministry of Education of China (Top 1 %)

Technical Skills

Programming Languages: Python, R, SQL, JavaScript, Html

Python Libraries for ML/DL: PyTorch, TensorFlow, Keras, Scikit-learn, SciPy, Numpy, Pandas, Geopandas

NLP: Scikit-learn, Genism, spaCy, Pyspark MLlib

Statistical Analysis: Time Series Analysis, Causal Inference, Bayesian Inference, Network Analysis, Experiment Design and Analysis

Distributed Computing: Spark, Pyspark, MapReduce, Hive, Yarn

Spatial Analysis & Simulation: ArcGIS, QGIS, GeoDa, SUMO, MATSim

Data Visualization: Tableau, Power BI, Vega, D3.js, CSS